National University of Computer and Emerging Sciences

FAST School of Computing **Fall-2023** **Islamabad Campus**

MT-2005: Probability and Statistics AI & DS

Sections (A,B,C,D)

Serial No:

2nd Sessional Exam

Total Time: 1 Hour

Total Marks: 30

Saturday, 4th November, 2023

Course Instructors

Hazber Samson, Ammara Ali, Waqas Munir

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Signature of Invigilator

Student Name **Roll No.** **Course Section** **Student Signature**

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt on question paper. Attempt all of them. Read the question carefully, understand the question, and then attempt it.
2. No additional sheet will be provided for rough work. Use the back of the last page for rough work.
3. If you need more space, write on the back side of the paper and clearly mark question and part number etc.
4. After asked to commence the exam, please verify that you have **five (5)** different printed pages including this title page. There are total of **4** questions.
5. Calculator sharing is strictly prohibited.
6. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.

	Q-1	Q-2	Q-3	Q-4	Total
Marks Obtained					
Total Marks	7	7	8	8	30

Question 1 [3+4=7 Marks]

(a) Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated
(a) with no restrictions?
(b) If each couple is to sit together?
(c) If all the men sit together to the right of all the women?

Solution

(a) $8! = 40320$

(b) C1 C2 C3 C4

$$arrangements = (2!\times 2!\times 2!\times 2!)\times 4! = 384$$

(c) W W W W M M M M

$$arrangements = 4!\times 4! = 576$$

(b) A certain carton of eggs has 3 bad eggs and 9 good eggs. An omelet is made of 3 eggs randomly chosen from the carton. What is the probability that omelet contains
(a) at least one bad egg.
(b) at most one bad egg.

Solution

Good Eggs = 9
Bad Eggs = 3
Total = 12
Select = 3

(a) $P(atleast\ one\ bad\ egg) = \dfrac{{}^{3}C_1 \times {}^{9}C_2 + {}^{3}C_2 \times {}^{9}C_1 + {}^{3}C_3 \times {}^{9}C_0}{{}^{12}C_3} = \dfrac{34}{55}$

(a) $P(atmost\ one\ bad\ egg) = \dfrac{{}^{3}C_1 \times {}^{9}C_2 + {}^{3}C_0 \times {}^{9}C_3}{{}^{12}C_3} = \dfrac{48}{55}$

Question 2 [3+4=7 Marks]

(a) In a certain federal prison, it is known that 2/3 of the inmates are under 25 years of age. It is also known that 3/5 of the inmates are male and that 5/8 of the inmates are female or 25 years of age or older. What is the probability that a prisoner selected at random from this prison is female and at least 25 years old?

Solution

U: Under 25
M: Male
F: Female
$P(F \cap U^c) = ?$

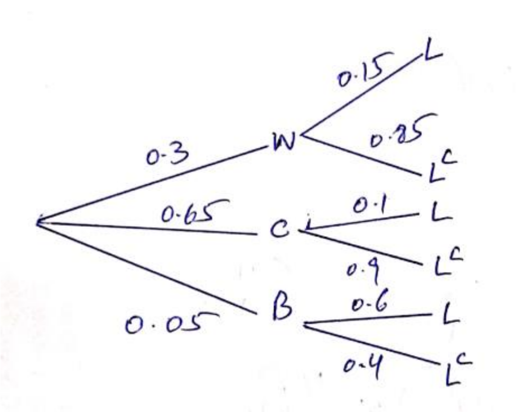
Using Addition Law of Probability $P(F \cup U^c) = P(F) + P(U^c) - P(F \cap U^c)$

$$\frac{5}{8} = (1 - \frac{3}{5}) + (1 - \frac{2}{3}) - P(F \cap U^c)$$

$$P(F \cap U^c) = \frac{2}{5} + \frac{1}{3} - \frac{5}{8} = \frac{13}{120}$$

(b) When Ayesha goes to school, the probability that she walks is 0.3 and the probability that she cycles is 0.65. If she does not walk or cycle she takes the bus. When Ayesha walks the probability that she is late is 0.15. when she cycles the probability that she is late is 0.1 and when she takes bus the probability that she is late is 0.6. Given that Ayesha is late, find the probability that she cycles.

Solution



$$P(C|L) = \frac{P(C \cap L)}{P(L)} = \frac{(0.65)(0.1)}{(0.3)(0.15) + (0.65)(0.1) + (0.05)(0.6)} = 0.4642$$

Question 3 [4+4=8 Marks]

(a) There are 90 applicants for a job with the news department of a television station. Some of them are college graduates and some are not; some of them have at least three years' experience and some have not, with the exact breakdown being

	College graduates	*Not college graduates*
At least three years' experience	18	9
Less than three years' experience	36	27

If the order in which the applicants are interviewed by the station manager is random, G is the event that the first applicant interviewed is a college graduate, and T is the event that the first applicant interviewed has at least three years' experience, determine each of the following probabilities directly from the entries and the row and column totals of the table:

$(a)\ P(T|G)$

$(b)\ P(G^c|T^c)$

Solution

$$(a)\ P(T|G) = \frac{P(T \cap G)}{P(G)} = \frac{18/90}{54/90} = \frac{1}{3}$$

$$(b)\ P(G^c|T^c) = \frac{P(G^c \cap T^c)}{P(T^c)} = \frac{27/90}{63/90} = \frac{3}{7}$$

(b) A firm is accustomed to training operators who do certain tasks on a production line. Those operators who attend the training course are known to be able to meet their production quotas 90% of the time. New operators who do not take the training course only meet their quotas 65% of the time. Fifty percent of new operators attend the course. Given that a new operator meets her production quota, what is the probability that she attended the program?

Solution

Consider the events:
T: an operator is trained, $P(T) = 0.5$,
M an operator meets quota, $P(M \mid T) = 0.9$ and $P(M \mid T') = 0.65$.
$P(T \mid M) = \frac{P(M \mid T)P(T)}{P(M \mid T)P(T)+P(M \mid T')P(T')} = \frac{(0.9)(0.5)}{(0.9)(0.5)+(0.65)(0.5)} = 0.581.$

Question 4 [2+2+2+2=8 Marks]

A vegetable basket contains 12 peppers of which 3 are red, 4 are green and 5 are yellow. Three peppers are taken at random without replacement, from the basket.

(a) Let X denote number of green peppers, find probability distribution of X.

(b) Find distribution function for X also find $P(X \le 2)$

(c) Find $E(X)$ *and* $V(X)$

(d) Find $E(2X+3)$ *and* $V(3X+5)$

Solution

Total = 12
Red = 3
Green = 4
Yellow = 5

(a)

X	P(X)	F(X)	X P(X)	X2 P(X)
0	$\frac{{}^4C_0 \times {}^8C_3}{{}^{12}C_3} = \frac{14}{55}$	$\frac{14}{55}$	0	0
1	$\frac{{}^4C_1 \times {}^8C_2}{{}^{12}C_3} = \frac{28}{55}$	$\frac{42}{55}$	$\frac{28}{55}$	$\frac{28}{55}$
2	$\frac{{}^4C_2 \times {}^8C_{13}}{{}^{12}C_3} = \frac{12}{55}$	$\frac{54}{55}$	$\frac{24}{55}$	$\frac{48}{55}$
3	$\frac{{}^4C_{30} \times {}^8C_0}{{}^{12}C_3} = \frac{1}{55}$	$\frac{55}{55}$	$\frac{3}{55}$	$\frac{9}{55}$
SUM	1		1	17/11

(b) $P(X \le 2) = F(2) = 54/55$

(c) $E(X) = 1$ *and* $V(X) = 17/11 - 1 = 6/11 = 0.54$

(d) $E(2X+3) = 2E(X) + 3 = 2(1) + 3 = 5$ *and* $V(3X+5) = 9V(X) = 9(6/11) = 54/11 = 4.91$